

ARIMA Time Series Models in STATA

Opening Question

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Stationarity tells us that we can do so for a time series.

3 Key Assumptions for Stationarity

The expectation of variable X is not a function of time

$$E(X_t) = E(X_1) = E(X_2) = \dots = E(X) = \mu \quad (1)$$

The variance of X is not a function of time

$$\text{Var}(X_t) = \text{Var}(X_1) = \text{Var}(X_2) = \dots = \text{Var}(X) = \sigma^2 \quad (2)$$

Covariance depends only on the lag

$$\text{Cov}(X_t, X_{t-k}) = \gamma_k \quad (3)$$

Stationarity Conditions with Closing Price

After First Differencing:

- Mean stationarity | ✓
- Variance stationarity | ✗
- Covariance stationarity | ✓

- **Mean stationarity:** We can model the mean dynamics using a stable relationship over time.
- **Covariance stationarity:** Relationships between observations at different lags are stable over time, allowing us to use standard time-series modelling techniques.
- **No variance stationarity:** The variability of the process changes over time.
 - Constant-variance assumptions are not appropriate.
 - Standard errors should therefore be made robust to time-varying volatility.

What do we want to model?

Mean Dynamics

- Mean, Covariance Stationarity Crucial
- ARIMA family models are good candidates to model dynamics
 - AR, ARMA, ARIMA (X), SARIMA, SARIMAX

Variance Dynamics

- Stable Conditional Variance Dynamics
- Volatility Models are good candidates to model dynamics
 - ARCH, GARCH, EGARCH, GJR-GARCH, APARCH

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Note that if both mean and variance are of interest one can combine the mean models with volatility models (ARIMA + GARCH for example).

Implementing Time Series Models in Stata

Refer to your lesson 2 Arima do file

AR: Why Must the Coefficients Be Bounded?

Consider the AR(1) model:

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t$$

Suppose we receive a one-unit shock today.

Effect today = 1

Effect next period = ϕ

Effect two periods later = ϕ^2

Therefore, the effect of the shock over time is:

$$1, \quad \phi, \quad \phi^2, \quad \phi^3, \quad \dots$$

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Effect of the shock over time is:

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For the shock to eventually disappear:

$$|\phi| < 1$$

The smaller the coefficient, the faster shocks die out.